ECE 329 Exam 1 Review

Prof. Shao



Timeline






Lorentz force

Gauss's Law

Static Charge Densities

Div. & Curl

Conductors

Poisson and Laplace Equ.

Boundary conditions

Curl free field & V

Dielectrics

Summary of Maxwell's Equations

Faraday's Law	$\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$	$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$
Ampere's Law	$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$	$\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$
Gauss' Law	$\oiint_S \vec{B} \bullet d\vec{S} = 0$	$\nabla \bullet \vec{B} = 0$
Gauss' Law	$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$	$\nabla \bullet \vec{D} = \rho$
Continuity Eq.	$\oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$	$\nabla \bullet \vec{J} = -\frac{d\rho}{dt}$

Lorentz Force Equation

So if a region of space contains BOTH an **E** field and a **B** field, a moving charge will experience force from both at the same time…

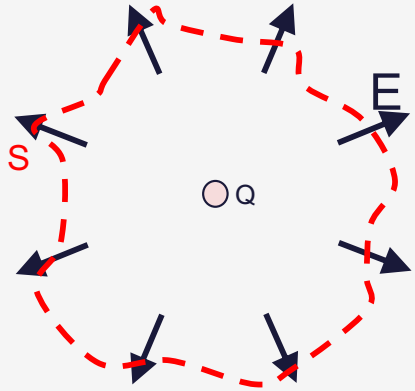
$$\vec{F}_{TOTAL} = \vec{F}_E + \vec{F}_M$$

$$\vec{F}_{TOTAL} = q\vec{E} + q\vec{v} \times \vec{B}$$

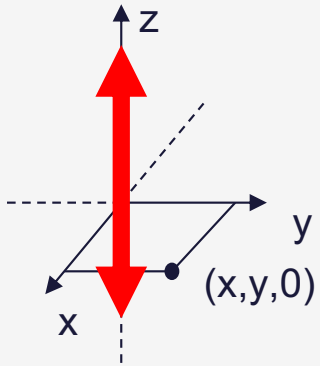
Gauss's Law

$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

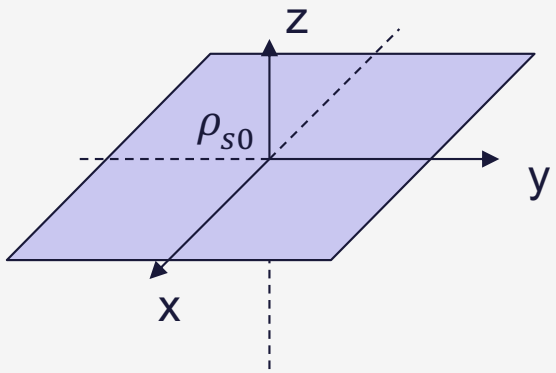
FLUX OUT = CHARGE ENCLOSED



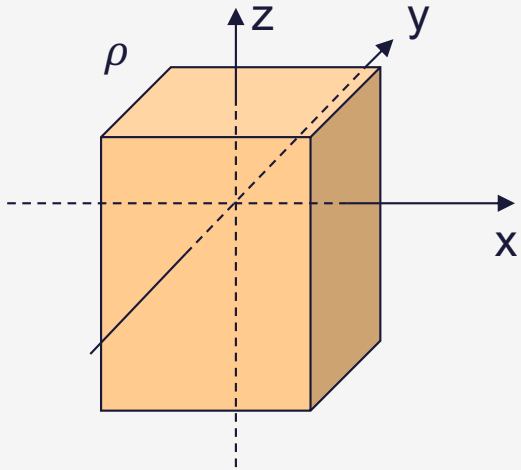
Point charge ρ (C)



Linear charge distribution ρ_l (C/m)



Surface charge density ρ_s (C/m^2)



Volume charge density ρ (C/m^3)

Charge use Impulse Notation

point charges at (x_0, y_0, z_0)	$\rho(x,y,z) = \rho\delta(x-x_0)\delta(y-y_0)\delta(z-z_0)$	ρ: C
line charges at $x = x_0$ and $y = y_0$	$\rho(x,y,z) = \rho_l\delta(x-x_0)(y-y_0)$	ρ_l: C/m
surface charges at $x = x_0$	$\rho(x,y,z) = \rho_s\delta(x-x_0)$	ρ_s : C/m^2

Summary of electrostatic fields

- Coulomb field

$$\mathbf{E} = \hat{r}\frac{Q}{4\pi\epsilon_o r^2} \text{ of a point charge } Q,$$

- Field

$$\mathbf{E} = \hat{r}\frac{\lambda}{2\pi\epsilon_o r} \text{ of constant line density } \lambda,$$

- Field

$$\mathbf{E} = \hat{x}\frac{\rho_s}{2\epsilon_o}\text{sgn}(x) \text{ of constant surface density } \rho_s,$$

- Field

$$\mathbf{E} = \hat{x}\frac{\rho x}{\epsilon_o} \text{ of constant volume density } \rho$$

Field Superposition

Electrostatics (no time varying)

The following are therefore equivalent:

v is conservative

$$\oint_C \vec{v} \bullet d\vec{l} = 0$$

v is curl-free

$$\int_a^b \vec{v} \bullet d\vec{l} \text{ is path independent}$$

$$\vec{v} = -\nabla f \text{ for some scalar potential f}$$

Del ∇ Operator

Del operator $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \hat{a}_x \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) - \hat{a}_y \left(\frac{\partial v_z}{\partial x} - \frac{\partial v_x}{\partial z} \right) + \hat{a}_z \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

$$\nabla \bullet \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{a}_x + \frac{\partial f}{\partial y} \hat{a}_y + \frac{\partial f}{\partial z} \hat{a}_z$$

Del **Operator** (Cartesian Coordinates) used to do gradient, divergence, and curl

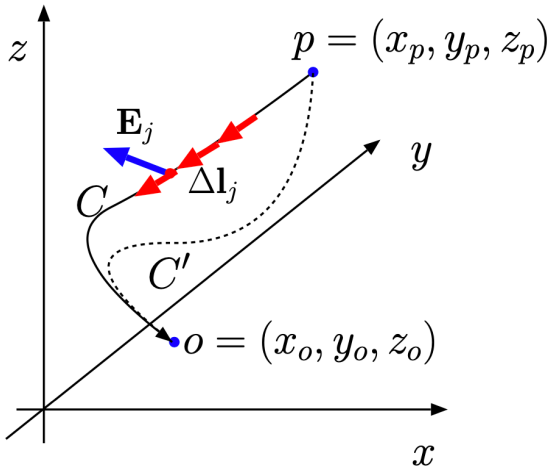
Electrostatic Potential V and electric field E

From V to E

$$\vec{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z\right)$$

From E to V

$$V = -\int_o^P \vec{E}\cdot d\vec{l} = \int_P^o \vec{E}\cdot d\vec{l}$$



$$d\mathbf{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz = (dx, dy, dz)$$

$$\mathbf{E}\cdot d\mathbf{l} = (E_x, E_y, E_z)\cdot(dx, dy, dz) = E_x dx + E_y dy + E_z dz.$$

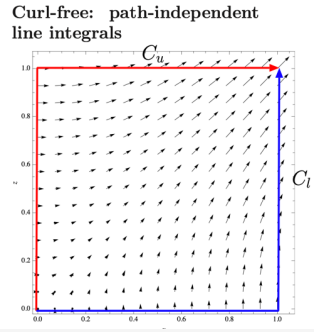
$$\int_p^o \mathbf{E}\cdot d\mathbf{l} = \int_p^o (E_x dx + E_y dy + E_z dz)$$

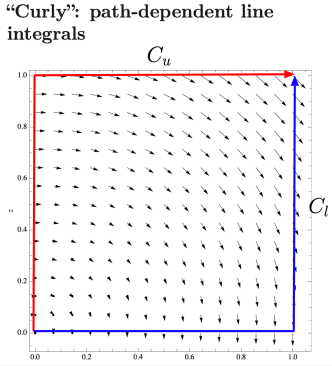
integration will in general be path dependent except for when E is curl-free

Path Independent and Curl Free (cont.)

The field $\vec{\boldsymbol{E}} = \hat{x}y \pm \hat{y}x$ is curl-free with the + sign, but not with − sign.
Calculate the line integral of $\vec{\boldsymbol{E}}$ (for both signs, ±) from a point $o = (0,0,0)$ to point $p = (1,1,0)$ for two different paths.

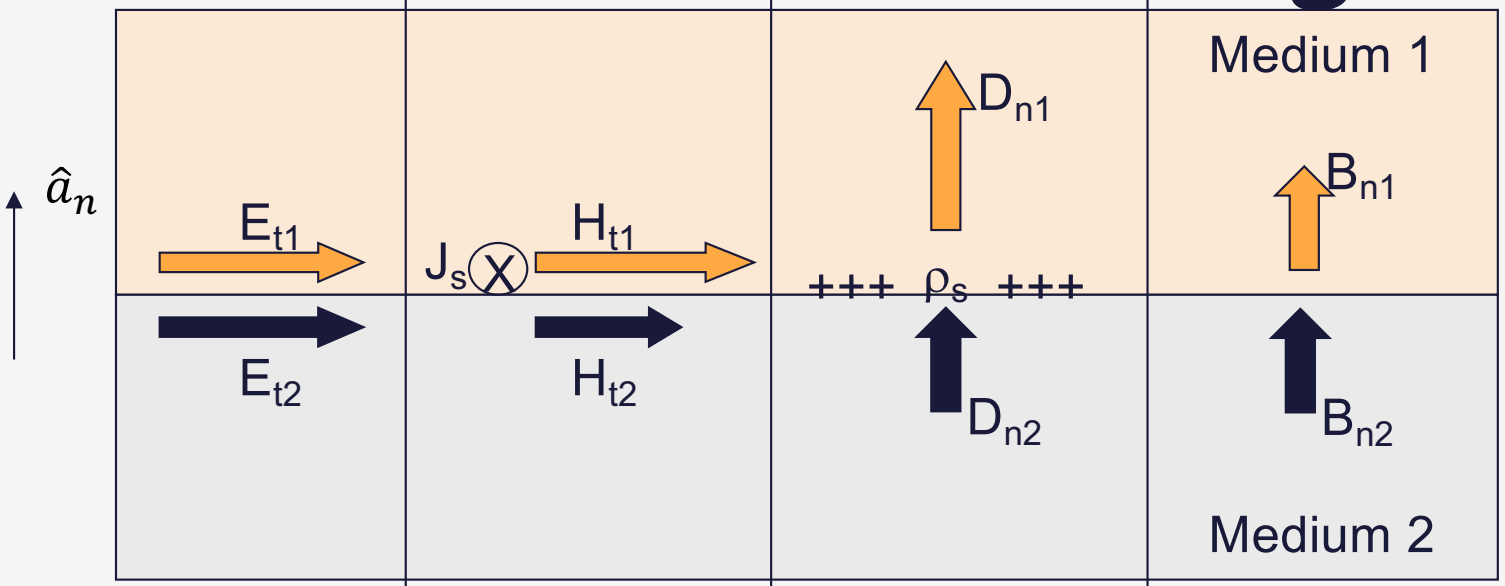
$$\int_o^p \mathbf{E} \cdot d\mathbf{l} = \int_o^p (E_x dx + E_y dy + E_z dz) = \int_o^p (y\,dx \pm x\,dy)$$





Example 2 in Prof. Kudeki's 329lect05.pdf

Remember this drawing!!



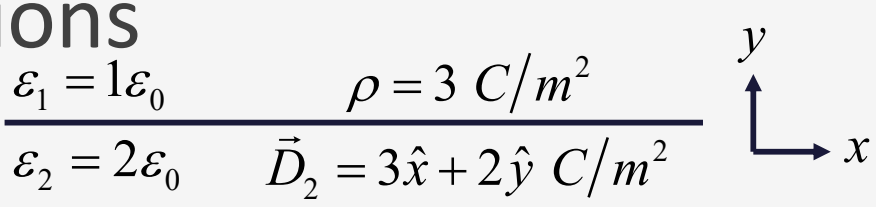
$E_{t1} - E_{t2} = 0$ $\quad \vec{H}_{t1} - \vec{H}_{t2} = \vec{J}_s$ $\quad D_{n1} - D_{n2} = \rho_s$ $\quad B_{n1} - B_{n2} = 0$

$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0$ $\quad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$ $\quad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$ $\quad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$

Challenge Question:

Boundary Conditions

- What is **D_1**?



$(a)\ 3\hat{x} + 2\hat{y}\ C/m^2$

$(b)\ 3\hat{x} + 5\hat{y}\ C/m^2$

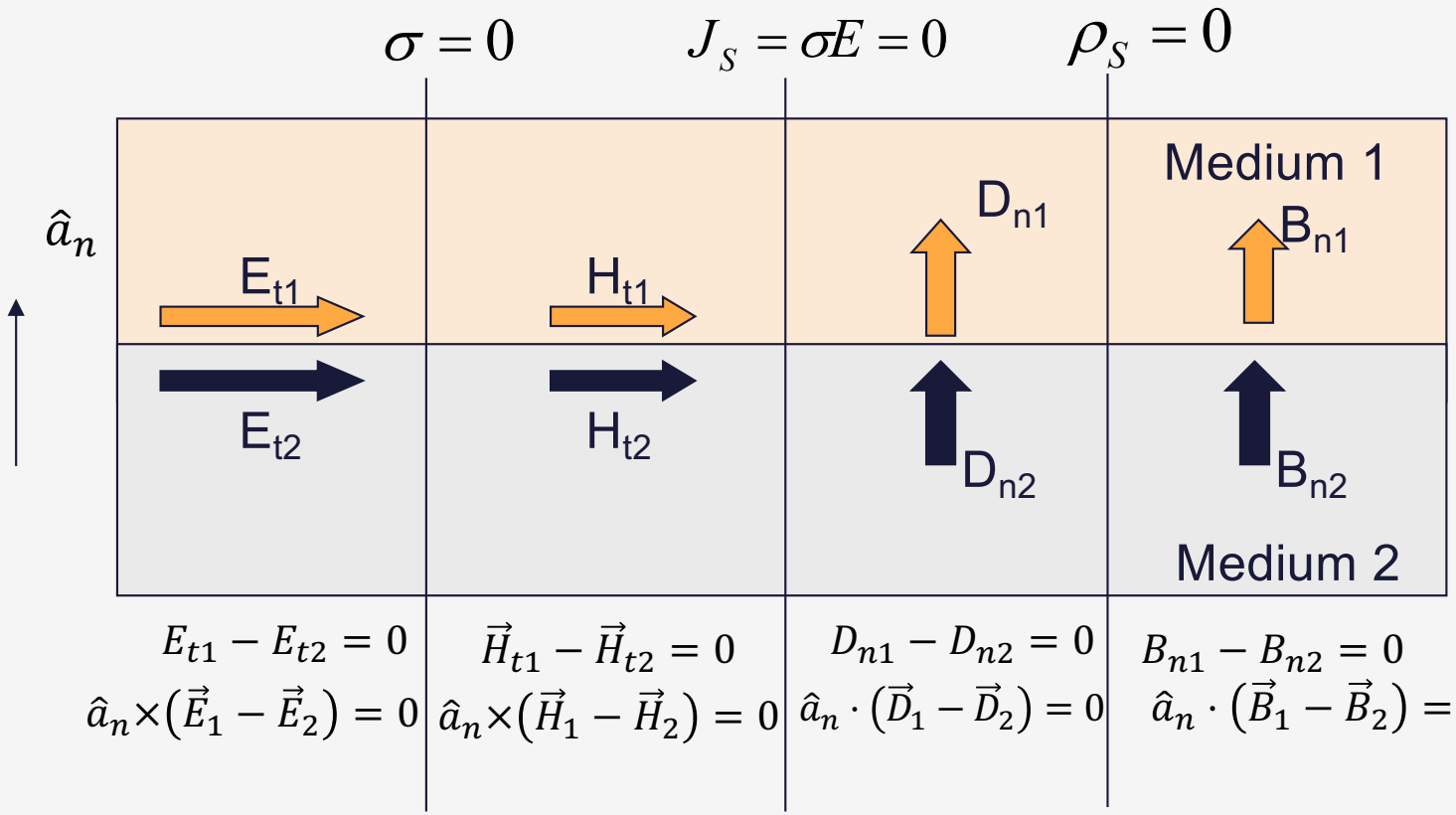
$(c)\ 3\hat{x} - 1\hat{y}\ C/m^2$

$(d)\ \frac{3}{2}\hat{x} + 5\hat{y}\ C/m^2$

$(e)\ \frac{3}{2}\hat{x} - 1\hat{y}\ C/m^2$

Example from Lect6-7_BoundaryCondition_Poisson.pdf

Boundary between Two Perfect Dielectrics



$E_{t1} - E_{t2} = 0$ | $\vec{H}_{t1} - \vec{H}_{t2} = 0$ | $D_{n1} - D_{n2} = 0$ | $B_{n1} - B_{n2} = 0$

$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0$ | $\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = 0$ | $\hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = 0$ | $\hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$

Poisson Equation for Potentials (static fields, constant ε)

$$\nabla^2\Phi = -\frac{\rho}{\varepsilon}$$

$$\nabla^2\Phi = \frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} + \frac{\partial^2\Phi}{\partial z^2}$$

Scalar

Known as “Poisson’s Equation”

It’s just Gauss’ Law in terms of Potentials for a Static Field

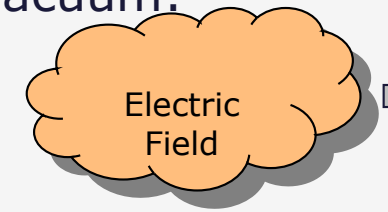
If ρ=0, it is known as “Laplace’s Equation”

Conductors

- Ohm's Law: _____
- Conductivity: $\sigma = \frac{1}{\text{Resistivity}} = \frac{l}{R \cdot A}$
- Inside a perfect conductor, $\vec{E} =$?

Displacement field $\vec{D}$

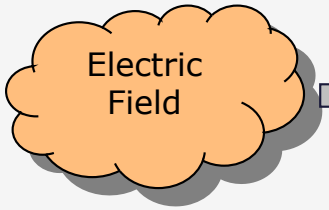
In vacuum:



Displacement (electric flux density)

$$\vec{D} = \epsilon_o \vec{\mathbf{E}}$$

In dielectric:



Displacement (electric flux density)

$$\vec{D} = \epsilon_o \vec{\mathbf{E}} + \vec{P}$$

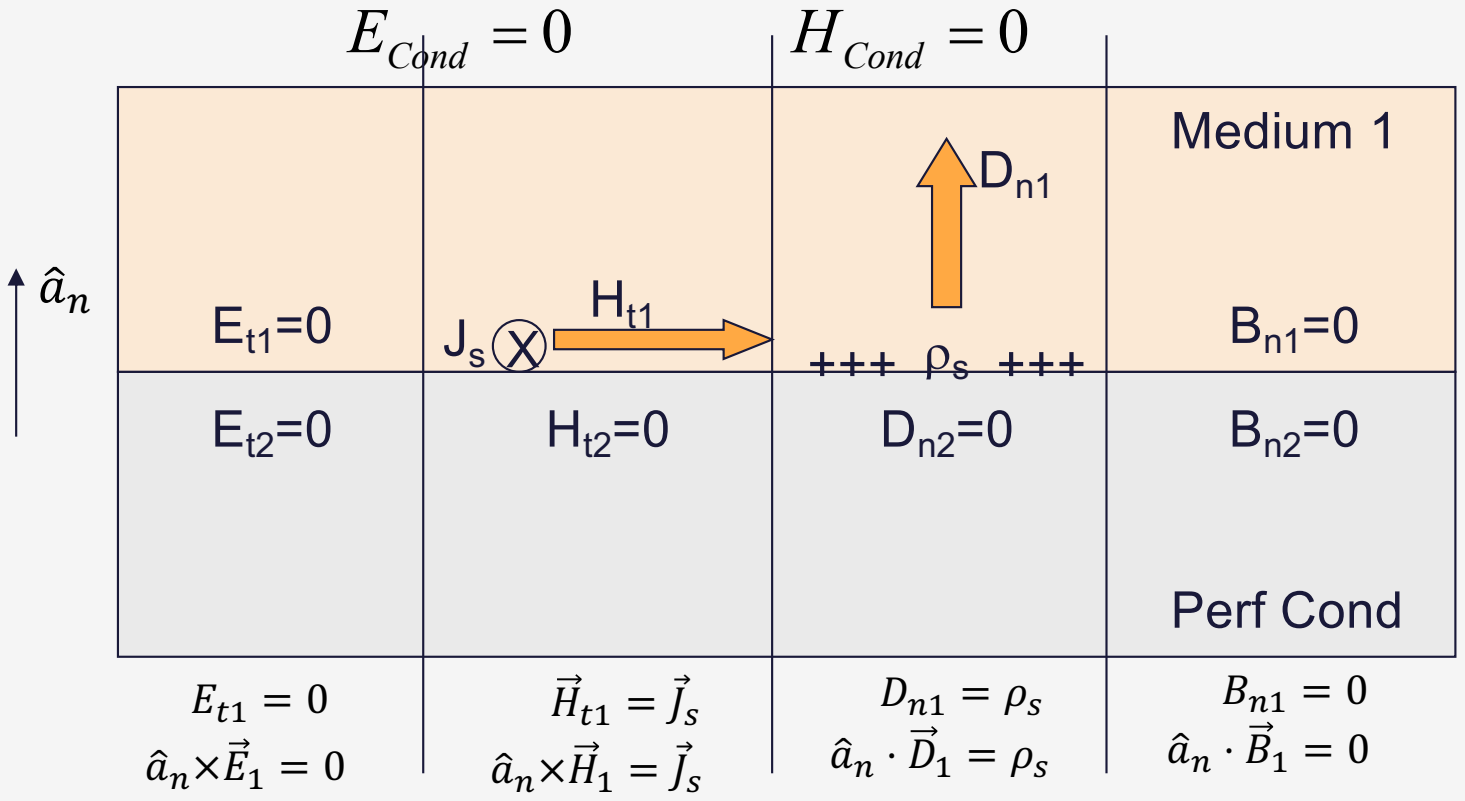
Item $\epsilon_o \vec{\mathbf{E}}$: is the same as the case in vacuum

Item "$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$": is the polarization due to contribution of dipole formation in dielectric under applied $\vec{\mathbf{E}}$ field

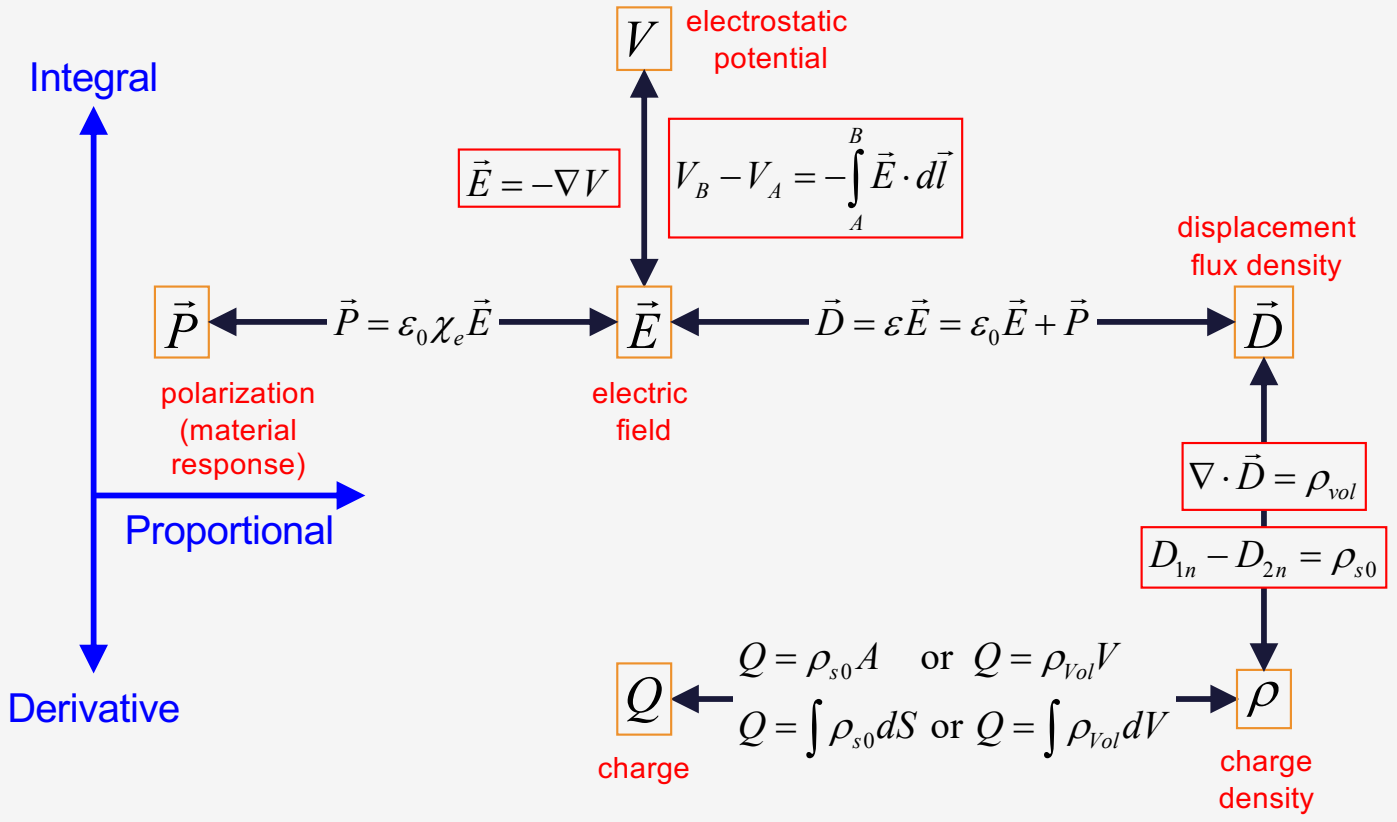
$$\varepsilon_r = (1 + \chi_e)$$
$$\varepsilon = \varepsilon_0 \varepsilon_r$$

$$\vec{D} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E} = \epsilon_o \vec{\mathbf{E}} + \vec{P}$$

Surface of Perfect Conductor



Connection of Concepts for Electrostatics



Thanks!

Wish you luck in Exam 1!

